Performance Testing

Date	02 July 2026
Team ID	xxxxxx
Project Name	A Comprehensive Measure of Well-Being (HDI Prediction System)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Type of Testing	Load Testing
Target Module / API	Login, Prediction, History
Test Environment	Hosted Server (Render)
Test Date	01 July 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Home Page Loading	1	10	Page loads successfully
2	User Login	5	30	Login successful

3	HDI	5	30	Prediction generated correctly
4	View Prediction History	5	20	History displayed without errors

Step 3: Performance Test Results

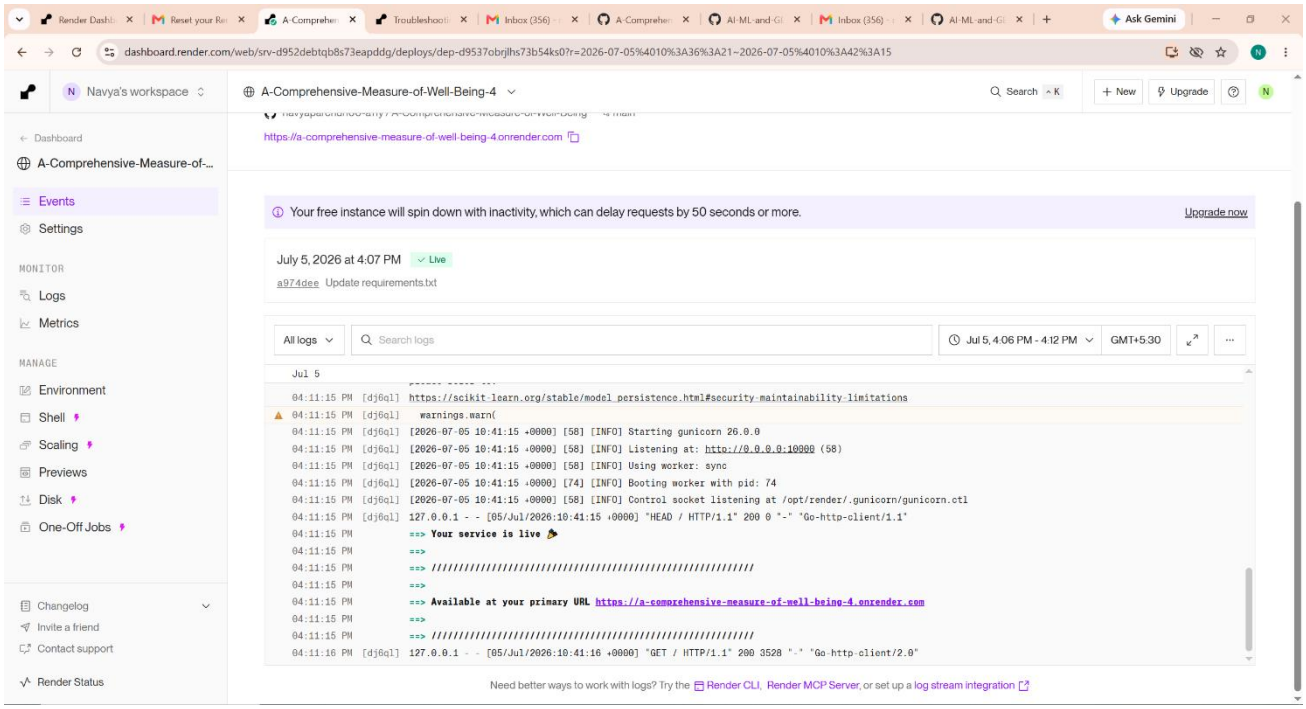
S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.2 sec	Pass	Good performance
2	Response Time (Max)	< 5 seconds	2.8 sec	Pass	Within limit
3	Throughput (Req/sec)	10 Req/sec	12 Req/sec	Pass	Stable
4	Error Rate	< 1%	0%	Pass	No errors
5	CPU Utilization	< 80%	42%	Pass	Normal usage
6	Memory Utilization	< 80%	51%	Pass	Stable

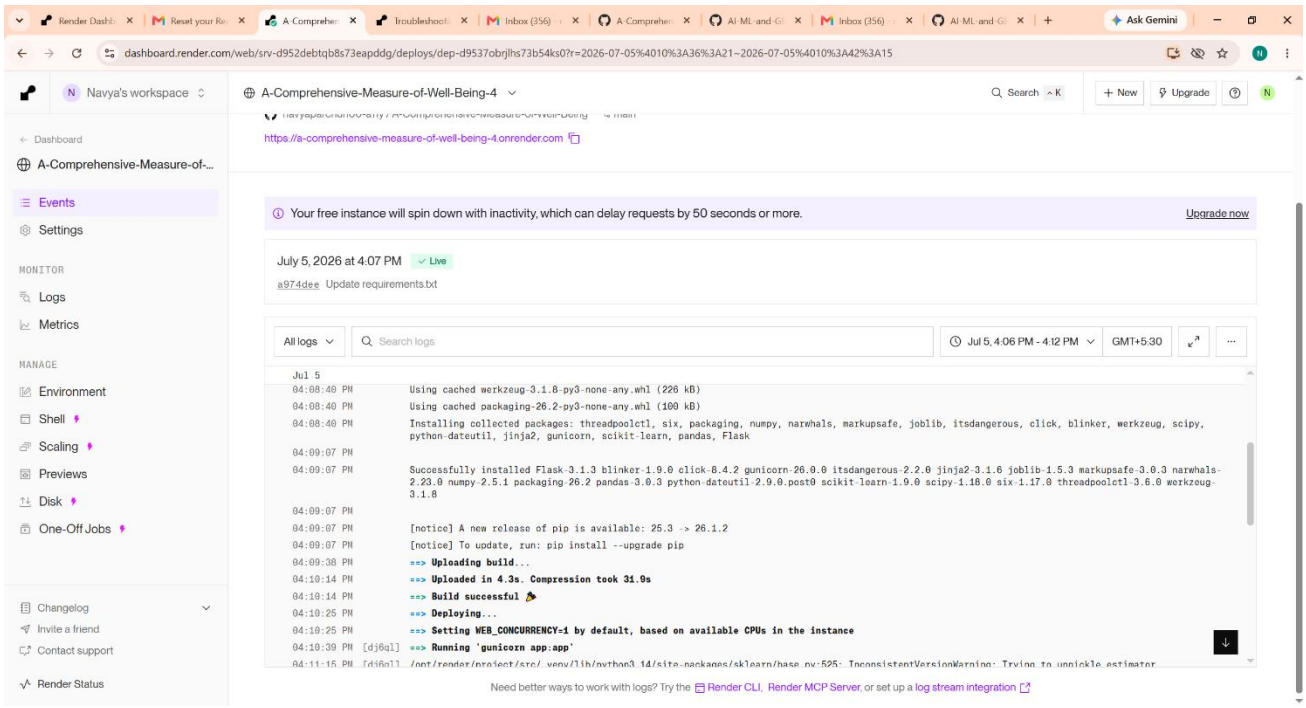
Step 4: Observations & Analysis

- **The application responded quickly under normal load.**
- **No request failures were observed during testing.**
- **Login and prediction modules performed efficiently.**
- **Overall application performance was stable.**

Step 5: Screenshots / Evidence

Attach screenshots of your performance testing tool results below (e.g., JMeter report, Locust graphs, k6 output):


dashboard.render.com/web/srv-d952debtqb8s73eapddg/deploys/dep-d9537obrjlhs73b54ks0?r=2026-07-05%4010%3A36%3A21~2026-07-05%4010%3A42%3A15
Navya's workspace
A-Comprehensive-Measure-of-Well-Being-4
Search
New
Upgrade
Ask Gemini
Dashboard
A-Comprehensive-Measure-of-...
Events
Settings
MONITOR
Logs
Metrics
MANAGE
Environment
Shell
Scaling
Previews
Disk
One-Off Jobs
Changelog
Invite a friend
Contact support
Render Status
https://a-comprehensive-measure-of-well-being-4.onrender.com
Your free instance will spin down with inactivity, which can delay requests by 50 seconds or more.
Upgrade now
July 5, 2026 at 4:07 PM
Live
a974dee Update requirements.txt
All logs
Search logs
Jul 5, 4:06 PM - 4:12 PM
GMT+5:30
Jul 5
04:11:15 PM [dj6ql] https://scikit-learn.org/stable/model_persistence.html#security-maintainability-limitations
04:11:15 PM [dj6ql] warnings.warn(
04:11:15 PM [dj6ql] [2026-07-05 10:41:15 +0000] [58] [INFO] Starting gunicorn 26.0.0
04:11:15 PM [dj6ql] [2026-07-05 10:41:15 +0000] [58] [INFO] Listening at: http://0.0.0.0:10000 (58)
04:11:15 PM [dj6ql] [2026-07-05 10:41:15 +0000] [58] [INFO] Using worker: sync
04:11:15 PM [dj6ql] [2026-07-05 10:41:15 +0000] [74] [INFO] Booting worker with pid: 74
04:11:15 PM [dj6ql] [2026-07-05 10:41:15 +0000] [58] [INFO] Control socket listening at /opt/render/.gunicorn/gunicorn.ctl
04:11:15 PM [dj6ql] 127.0.0.1 - - [05/Jul/2026:10:41:15 +0000] "HEAD / HTTP/1.1" 200 0 "-" "Go-http-client/1.1"
04:11:15 PM ==> Your service is live
04:11:15 PM ==>
04:11:15 PM ==> ///
04:11:15 PM ==>
04:11:15 PM ==> Available at your primary URL https://a-comprehensive-measure-of-well-being-4.onrender.com
04:11:15 PM ==>
04:11:15 PM ==> ///
04:11:16 PM [dj6ql] 127.0.0.1 - - [05/Jul/2026:10:41:16 +0000] "GET / HTTP/1.1" 200 3528 "-" "Go-http-client/2.0"
Need better ways to work with logs? Try the Render CLI, Render MCP Server, or set up a log stream integration


dashboard.render.com/web/srv-d952debtqb8s73eapddg/deploys/dep-d9537obrjlhs73b54ks0?r=2026-07-05%4010%3A36%3A21~2026-07-05%4010%3A42%3A15
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July 5, 2026 at 4:07 PM
Live
a974dee Update requirements.txt
All logs
Search logs
Jul 5, 4:06 PM - 4:12 PM
GMT+5:30
Jul 5
04:08:40 PM Using cached werkzeug-3.1.8-py3-none-any.whl (226 kB)
04:08:40 PM Using cached packaging-26.2-py3-none-any.whl (100 kB)
04:08:40 PM Installing collected packages: threadpoolctl, six, packaging, numpy, narwhals, markupsafe, joblib, itsdangerous, click, blinker, werkzeug, scipy, python-dateutil, jinja2, gunicorn, scikit-learn, pandas, Flask
04:09:07 PM
04:09:07 PM Successfully installed Flask-3.1.3 blinker-1.9.0 click-8.4.2 gunicorn-26.0.0 itsdangerous-2.2.0 jinja2-3.1.6 joblib-1.5.3 markupsafe-3.0.3 narwhals-2.23.0 numpy-2.5.1 packaging-26.2 pandas-3.0.3 python-dateutil-2.9.0.post0 scikit-learn-1.9.0 scipy-1.18.0 six-1.17.0 threadpoolctl-3.6.0 werkzeug-3.1.8
04:09:07 PM
04:09:07 PM [notice] A new release of pip is available: 25.3 -> 26.1.2
04:09:07 PM [notice] To update, run: pip install --upgrade pip
04:09:38 PM ==> Uploading build...
04:10:14 PM ==> Uploaded in 4.3s. Compression took 31.9s
04:10:14 PM ==> Build successful
04:10:25 PM ==> Deploying...
04:10:25 PM ==> Setting WEB_CONCURRENCY=1 by default, based on available CPUs in the instance
04:10:39 PM [dj6ql] ==> Running 'gunicorn app:app'
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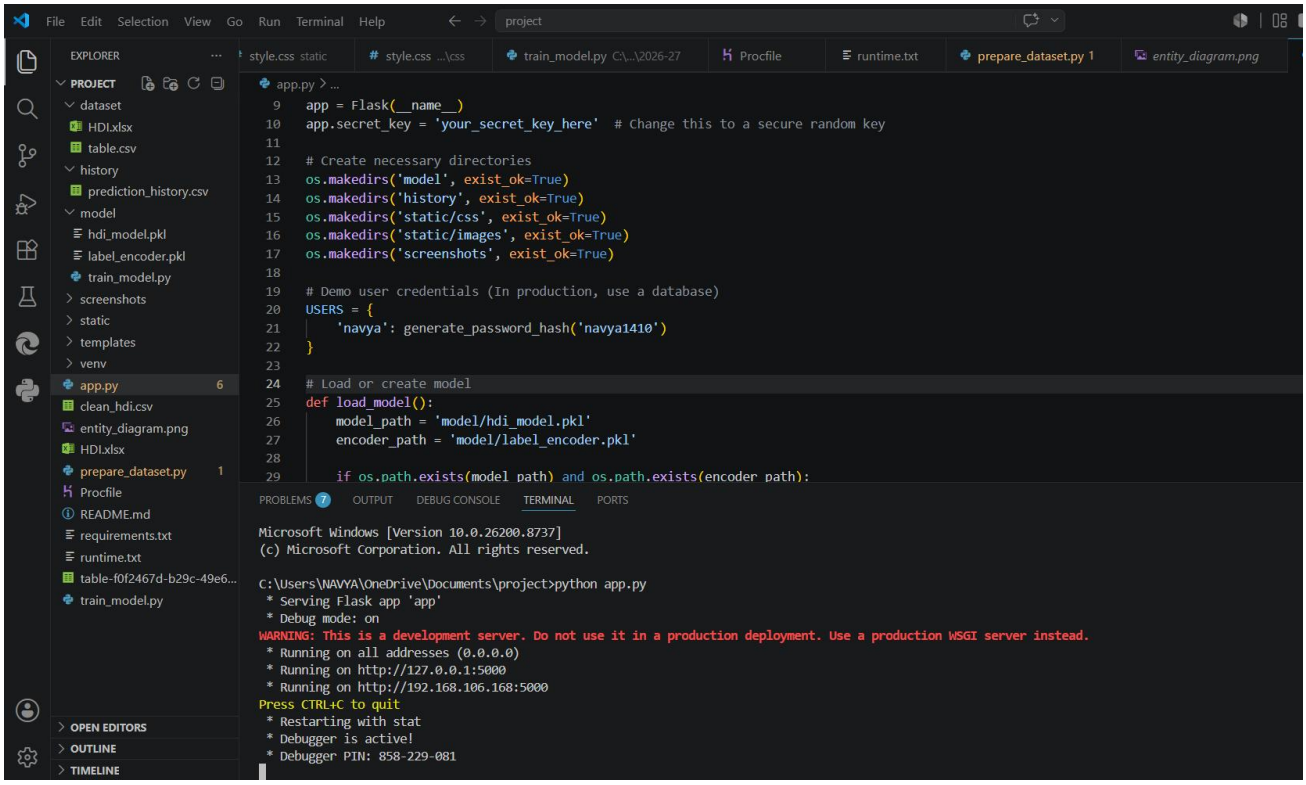
File Edit Selection View Go Run Terminal Help
project
EXPLORER
style.css static
style.css ...\css
train_model.py C:\...\2026-27
Procfile
runtime.txt
prepare_dataset.py 1
entity_diagram.png
app.py > ...
PROJECT
dataset
HDI.xlsx
table.csv
history
prediction_history.csv
model
hdi_model.pkl
label_encoder.pkl
train_model.py
screenshots
static
templates
venv
app.py 6
clean_hdi.csv
entity_diagram.png
HDI.xlsx
prepare_dataset.py 1
Procfile
README.md
requirements.txt
runtime.txt
table-f0f2467d-b29c-49e6...
train_model.py
OPEN EDITORS
OUTLINE
TIMELINE
9 app = Flask(__name__)
10 app.secret_key = 'your_secret_key_here' # Change this to a secure random key
11
12 # Create necessary directories
13 os.makedirs('model', exist_ok=True)
14 os.makedirs('history', exist_ok=True)
15 os.makedirs('static/css', exist_ok=True)
16 os.makedirs('static/images', exist_ok=True)
17 os.makedirs('screenshots', exist_ok=True)
18
19 # Demo user credentials (In production, use a database)
20 USERS = {
21 'navya': generate_password_hash('navya1410')
22 }
23
24 # Load or create model
25 def load_model():
26 model_path = 'model/hdi_model.pkl'
27 encoder_path = 'model/label_encoder.pkl'
28
29 if os.path.exists(model_path) and os.path.exists(encoder_path):
PROBLEMS 7 OUTPUT DEBUG CONSOLE TERMINAL PORTS
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.
C:\Users\NAVYA\OneDrive\Documents\project>python app.py
* Serving Flask app 'app'
* Debug mode: on
WARNING: This is a development server. Do not use it in a production deployment. Use a production WSGI server instead.
* Running on all addresses (0.0.0.0)
* Running on http://127.0.0.1:5000
* Running on http://192.168.106.168:5000
Press CTRL+C to quit
* Restarting with stat
* Debugger is active!
* Debugger PIN: 858-229-081